News Article Classification: A Linear Support Vector Machine Approach

D'Amico Davide, Rainone Gerardo
Politecnico di Torino
Student ids: s353778, s354059
s353778@studenti.polito.it, s354059@studenti.polito.it

***Abstract*—This report addresses the problem of automatically categorizing news articles into seven predefined topics using machine learning techniques. A classification pipeline was developed leveraging textual content processed via TF-IDF vectorization, alongside metadata features such as source and page rank. Several algorithms were evaluated, with Linear Support Vector Classification (LinearSVC) yielding the best performance. Through hyperparameter tuning, the final model achieved a Macro F1 score of approximately 0.72 on the validation set, demonstrating the effectiveness of high-dimensional sparse features for text classification tasks.**

I. Problem overview

The objective of this project is to develop a machine learning model capable of classifying news articles into one of seven categories: International, Business, Technology, Entertainment, Sports, General, and Health. The dataset provided consists of approximately 100,000 instances in the development set and a separate evaluation set for final predictions.

Each record contains textual data (title and article content) and metadata (source and page rank). An exploratory analysis of the development set reveals a significant class imbalance. As shown in Figure 1, Class 0 (International News) is the most frequent, appearing over 20,000 times, while classes such as 6 (Health) are drastically underrepresented. This imbalance poses a significant challenge for classification, as standard accuracy metrics would be biased toward the majority class. Consequently, the **Macro F1 score** was selected as the primary evaluation metric, as it calculates the harmonic mean of precision and recall for each class independently and then averages them, treating all classes equally regardless of their frequency.

II. Proposed approach

The proposed solution implements a supervised learning pipeline that integrates both textual and numerical features. The approach focuses on robust preprocessing to handle the high dimensionality of text data and rigorous model selection to identify the most effective classifier.

A. Preprocessing

Data preprocessing is crucial for transforming raw text into a format suitable for machine learning algorithms.

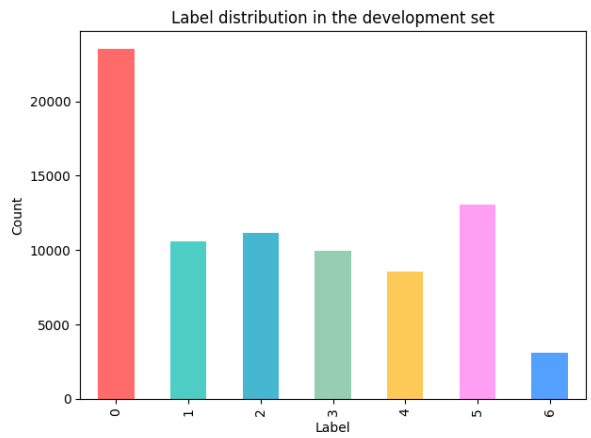

Fig. 1. Distribution of target labels in the development set. The dataset exhibits significant class imbalance, with Class 0 being dominant and Class 6 being the minority.

1) Feature Engineering: The textual features were constructed by concatenating the `title` and `article` fields into a single `text` column. This strategy ensures that the model can leverage keywords found in headlines, which are often highly descriptive, alongside the context provided by the full article body. Missing values in these fields were replaced with empty strings to ensure consistency during concatenation.

2) Vectorization and Scaling: The text data was transformed using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. This technique reduces the weight of common words that offer little discriminative power while highlighting terms unique to specific documents. English stop words were removed to reduce noise, and unigrams and bigrams ($ngram_range = (1, 2)$) were extracted to capture local context (e.g., "White House" vs. "White" and "House").

The metadata features were processed as follows:

- **Source:** The categorical `source` variable was encoded using One-Hot Encoding. This is necessary because the news source often correlates strongly with specific topics (e.g., "ESPN" implies Sports).
- **Page Rank:** The numerical `page_rank` feature was standardized using a Standard Scaler to ensure zero mean and unit variance, preventing it from dominating the objective function of distance-based models.

3) Vocabulary Optimization: A critical aspect of TF-IDF is managing the vocabulary size. Parameters `min_df` (minimum document frequency) and `max_df` (maximum document frequency) were tuned to remove rare and overly common terms. Figure 2 illustrates the impact of these parameters on vocabulary size. A configuration of `min_df=3` and `max_df=0.8` was found to offer an optimal balance, significantly reducing the feature space without losing discriminative information.

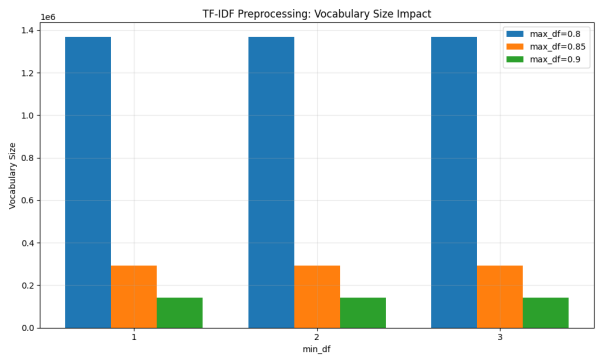

Fig. 2. Impact of `min_df` and `max_df` parameters on the vocabulary size generated by the TF-IDF vectorizer.

B. Model selection

Three distinct classification algorithms were evaluated using Stratified K-Fold cross-validation ($k = 3$) to ensure robustness and prevent overfitting. The algorithms selected represent different learning paradigms:

1) **LinearSVC:** A Support Vector Machine with a linear kernel, implemented via LibLinear. It is known for its efficiency in high-dimensional sparse spaces typical of text data.
2) **K-Nearest Neighbors (KNN):** A non-parametric method based on a distance metric (like Euclidean distance). ($k = 5$).
3) **Decision Tree:** A tree-based model trained with default parameters.

The results of the preliminary model selection are summarized in Table I.

TABLE I
COMPARISON OF CLASSIFICATION MODELS USING 3-FOLD CROSS-VALIDATION ON THE DEVELOPMENT SET.

Model	Mean F1 Macro	Std Dev
LinearSVC	**0.7118**	**0.0020**
KNN	0.6525	0.0011
Decision Tree	0.6101	0.0015

The LinearSVC outperformed the other models by a significant margin. Text classification tasks often result in high-dimensional feature spaces where linear decision boundaries are effective. Conversely, KNN suffered from the "curse of dimensionality," where distance measures become less meaningful in high-dimensional space. The single Decision Tree failed to capture the complex linguistic patterns as effectively as the SVM. Consequently, LinearSVC was selected for the final pipeline.

C. Hyperparameters tuning

To further optimize the LinearSVC model, a Grid Search was performed with 5-fold cross-validation. The search space included:

- **TF-IDF `min_df`:** $\{1, 2, 3\}$
- **TF-IDF `max_df`:** $\{0.8, 0.85, 0.9\}$
- **Regularization parameter** C**:** $\{0.2, 0.3, 0.4, 0.5\}$
- **Class Weight:** `balanced` (to address class imbalance)

The impact of the regularization parameter C on the Macro F1 score is depicted in Figure 3. Lower values of C (stronger regularization) generally yielded better generalization, with the peak performance observed at $C = 0.2$. This suggests that the model benefits from a wider margin to avoid overfitting the noisy text data.

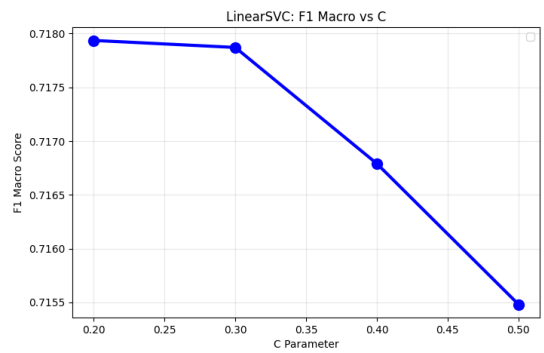

Fig. 3. Macro F1 score performance of the LinearSVC model across different values of the regularization parameter C. A clear downward trend indicates that stronger regularization (lower C) improves performance.

III. RESULTS

The final model was trained using the optimal hyperparameters identified via Grid Search: `min_df=3`, `max_df=0.8`, `C=0.2`, and `class_weight='balanced'`.

The model achieved a Macro F1 score of approximately **0.72** on the validation set. Figure 4 presents the confusion matrix for the predictions. The diagonal dominance indicates that the model correctly classifies the majority of instances across all classes.

However, some confusion persists between specific class pairs. For instance, there is noticeable misclassification between Class 0 (International) and Class 5 (General News), likely due to the semantic overlap in article content where international events are often categorized broadly. Similarly, Class 4 (Sports) and Class 5 show some degree of ambiguity. Despite these challenges, the inclusion of the `class_weight='balanced'` parameter ensured that minority classes (like Class 6) were not ignored, maintaining acceptable recall rates across the board.

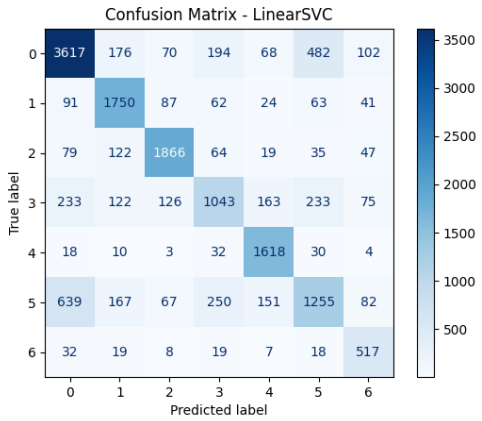

Fig. 4. Confusion Matrix of the optimized LinearSVC model. The integers on the axes represent the 7 target classes. Darker squares on the diagonal indicate correct predictions.

IV. Discussion

The proposed approach demonstrates that a linear model combined with TF-IDF vectorization is a strong baseline for news classification. The feature extraction process successfully captured the semantic differences between topics, and the use of metadata provided additional discriminative power.

Several observations can be drawn from the results:

- **Efficiency:** LinearSVC is computationally efficient, making it suitable for large datasets like the one provided. It scales linearly with the number of training samples.
- **Regularization:** The text data is sparse and high-dimensional; thus, strict regularization (low C) was necessary to prevent overfitting.
- **Limitations:** The current approach relies on bag-of-words representations (n-grams), which ignore the sequential nature of text. It cannot capture long-range dependencies or context beyond the immediate bigram window.

Future improvements could involve more advanced natural language processing techniques, such as word embeddings (e.g., Word2Vec) or transformer-based models (e.g., BERT), which could better capture semantic context and disambiguate complex sentences. Additionally, more sophisticated cleaning of the raw text could reduce noise and improve the quality of the TF-IDF.